

A CH obstruction to lifting automorphisms modulo locally null operators

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Source and target problem

The source paper is Piotr Koszmider and Cristobal Rodriguez-Porras, *On automorphisms of the Banach space ℓ_∞/c_0* , arXiv:1501.03466 [1].

In Problem 7.1, the authors ask whether it is consistent, possibly under PFA or OCA+MA, that every automorphism $T : \ell_\infty/c_0 \rightarrow \ell_\infty/c_0$ can be represented as

$$T = [R] + S,$$

where $R : \ell_\infty \rightarrow \ell_\infty$ induces an operator $[R]$ on the quotient and $S : \ell_\infty/c_0 \rightarrow \ell_\infty/c_0$ is locally null.

7. OPEN PROBLEMS AND FINAL REMARKS

In this section we mention some open problems and some observations related to them. This should not be considered as a full list of urgent open problems concerning the Banach space ℓ_∞/c_0 , for example we do not touch problems related to the primariness of ℓ_∞/c_0 (see [18], [17], [28]) subspaces of ℓ_∞/c_0 ([7], [29], [49], [30]), ℓ_∞ -sums ([18], [17], [8]) or extensions of operators on ℓ_∞/c_0 ([9], [4]).

Problem 7.1. *Is it consistent (does it follow from PFA or OCA+MA) that every automorphism $T : \ell_\infty/c_0 \rightarrow \ell_\infty/c_0$ can be lifted modulo a locally null operator? That is, is every such operator of the form $T = [R] + S$, where $R : \ell_\infty \rightarrow \ell_\infty$ and S is locally null?*

Source ingredients

The argument uses two results from the source paper. First, Corollary 4.9 states that every liftable isomorphic embedding of ℓ_∞/c_0 into itself is right-locally trivial.

Corollary 4.9. *Every liftable isomorphic embedding $T : \ell_\infty/c_0 \rightarrow \ell_\infty/c_0$ is right-locally trivial.*

Proof. Since T is liftable, there exist $R_0, R_1 : \ell_\infty \rightarrow \ell_\infty$ an antimatrix operator and one given by a c_0 -matrix, respectively, such that $T = [R_0 + R_1] = [R_0] + [R_1]$. Fix an infinite $A \subseteq \mathbb{N}$. By [4.7] take an infinite $A_0 \subseteq A$ such that $T \circ I_{A_0} = [R_1] \circ I_{A_0}$. Then, $[R_1] \circ I_{A_0}$ is a matrix operator which is an isomorphic embedding, so by [4.8] there exist infinite $A_1 \subseteq A_0$ and $B \subseteq \mathbb{N}$ such that $P_B \circ T \circ I_{A_1} = P_B \circ [R_1] \circ I_{A_1}$ is trivial. $\square$

Corollary 4.10. *Every liftable isomorphic embedding $T : \ell_\infty/c_0 \rightarrow \ell_\infty/c_0$ is right-locally an isomorphic matrix operator.*

Second, Theorem 6.9 states that, under CH, there is a nowhere trivial automorphism of ℓ_∞/c_0 .

Theorem 6.9 (CH). *There is an automorphism of ℓ_∞/c_0 with no fountains and no funnels which is nowhere trivial.*

Idea of the proof

The key point is that locally null errors vanish after passing to an infinite coordinate subspace. If $T = [R] + S$ and S is locally null, then on some $\ell_\infty(A)/c_0(A)$ we have $SI_A = 0$, hence $TI_A = [R]I_A$. Since T is an automorphism, this restriction of $[R]$ is a liftable isomorphic embedding. The source Corollary 4.9 then forces a further restriction to be trivial. Therefore T must be somewhere trivial.

Theorem 1. *Let $T : \ell_\infty/c_0 \rightarrow \ell_\infty/c_0$ be an automorphism. If*

$$T = [R] + S$$

for some quotient-induced operator $[R]$ and some locally null operator $S : \ell_\infty/c_0 \rightarrow \ell_\infty/c_0$, then T is somewhere trivial.

Proof. Let $I_A : \ell_\infty(A)/c_0(A) \rightarrow \ell_\infty/c_0$ and $P_B : \ell_\infty/c_0 \rightarrow \ell_\infty(B)/c_0(B)$ denote the standard coordinate inclusion and projection.

Since S is locally null, applying the definition to $A = \mathbb{N}$ gives an infinite $A_0 \subseteq \mathbb{N}$ such that

$$SI_{A_0} = 0.$$

Consequently

$$TI_{A_0} = [R]I_{A_0}.$$

Because T is an automorphism, TI_{A_0} is an isomorphic embedding of $\ell_\infty(A_0)/c_0(A_0)$ into ℓ_∞/c_0 . Therefore $[R]I_{A_0}$ is also an isomorphic embedding, and it is liftable by restriction of the same lifting R to the coordinates in A_0 .

Identify $\ell_\infty(A_0)/c_0(A_0)$ with ℓ_∞/c_0 through any bijection $\mathbb{N} \rightarrow A_0$. Corollary 4.9 of the source paper applies to this liftable isomorphic embedding. Hence there are infinite sets $A_1 \subseteq A_0$, $B \subseteq \mathbb{N}$, a bijection $\sigma : B \rightarrow A_1$, and a nonzero scalar $r \in \mathbb{R}$ such that

$$P_B[R]I_{A_1}([f]) = [rf \circ \sigma] \quad (f \in \ell_\infty(A_1)).$$

But $SI_{A_1} = 0$, so $P_B TI_{A_1} = P_B [R]I_{A_1}$. Thus T is somewhere trivial. $\square$

Corollary 1 (CH). *Assuming the continuum hypothesis, there is an automorphism $T : \ell_\infty/c_0 \rightarrow \ell_\infty/c_0$ which cannot be written as $T = [R] + S$ with S locally null.*

Proof. By Theorem 6.9 of the source paper, CH yields an automorphism $T : \ell_\infty/c_0 \rightarrow \ell_\infty/c_0$ which is nowhere trivial. The theorem above shows that any automorphism admitting a decomposition $T = [R] + S$ with S locally null must be somewhere trivial. Hence this CH automorphism admits no such decomposition. $\square$

Scope and limitations

This is a partial negative result. It does not answer whether PFA or OCA+MA implies the lifting-modulo-locally-null property. Instead it proves the ZFC implication

$$\text{liftable modulo locally null} \implies \text{somewhere trivial}$$

for automorphisms, and then combines this implication with the source paper’s CH nowhere-trivial automorphism.

Equivalently, a positive answer to Problem 7.1 under PFA or OCA+MA would imply a positive answer to the later somewhere-triviality problem for automorphisms under the same axiom.

Novelty check

On 2026-06-20, I searched for exact phrases around

`ell_infty/c_0 locally null automorphism lift modulo`

and around ℓ_∞/c_0 , “somewhere trivial”, and “locally null operator”. The only direct hit found for the exact target was the source arXiv paper.

Human review recommendation

Review this as a likely valid partial result. The main point to check is that the source Corollary 4.9 applies after replacing the domain by $\ell_\infty(A_0)/c_0(A_0)$, which is canonically isomorphic to ℓ_∞/c_0 after choosing a bijection $\mathbb{N} \rightarrow A_0$.

References

- [1] Piotr Koszmider and Cristobal Rodriguez-Porras, *On automorphisms of the Banach space ℓ_∞/c_0* , arXiv:1501.03466, 2015.